

Comparatively testing the effect of reality modality on spatial memory

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ABSTRACT

Virtual and augmented reality hold great potential for understanding spatial cognition. However, it is unclear what effect reality modality has on our perception and interaction with our spatial surroundings. Here, participants performed a spatial memory task using passthrough augmented reality in the real world and in a virtual environment reconstructed by scanning the real environment. We found no significant differences by reality modality for subjective measures such as reported immersion, difficulty, enjoyment and cyber-sickness, nor did we find objective differences in performance. These results suggest limited effects on spatial tasks, and are promising for transfer between virtual and augmented scenarios.

Keywords: Spatial cognition, Spatial memory, navigation, Virtual Reality, Augmented Reality, Reality modality.

Index Terms: Human centered computing, Perception, Virtual reality

1 INTRODUCTION

Remembering the spatial locations of items around us is a critical part of everyday life. For example, we need to encode where we placed our car keys or phone, and then recall that location when we need to use them. But how do we perform these tasks? How accurately can we perform them and under what conditions? In recent years a lot of the research into these questions has utilized virtual environments, as they avoid many challenges with physical setups such as logistics, safety, cost etc. [1]. However, it is not clear if these virtual environments are fully equivalent to performing spatial tasks in the real world.

Some often-cited challenges for comparing real and virtual spatial performance are the differences in movement - virtual tasks often have participants stationary, enabling them to move via keystrokes or teleportation. Even setups such as omnidirectional treadmills do not fully capture natural human walking [2,8]. Another common challenge is differences in interfaces, as often the task in the virtual world has additional or different interface affordances (e.g. pointing with a finger vs. a controller or target icon) than those offered when performing the task in the real world [2,8]. A third challenge is differences in the environment - often the virtual task is conducted in a different location such as an open room with padded walls. This difference in location may lead to differences in parameters such as smell, air flow and more [2,8]. A fourth challenge, which we focus upon here is reality modality - is there an inherent difference in behavior simply due to being in a virtual environment even if all else is identical?

Disentangling Reality Modality from the other three factors and their effects is difficult given that these tasks typically includes all four. Here, we designed a spatial memory task with virtual and augmented reality versions aiming at isolating this factor - participants will perform the same physical task in the same

physical space interacting via the same interface, controlling respectively for effects of environment, movement and interface, With the conditions differing only in reality modality. We then ask - will we find differences in subjective experience across reality modality? Will we find differences in objective performance?

2 METHODS

10 healthy volunteers participated in our study. All provided their informed consent and the study was approved by the IRB of Ben Gurion University. We created an adapted version of the "Treasure Hunt" spatial memory paradigm (Fig 1), previously used for researching spatial memory behavior [4-6], and its neural basis with intracranial recordings [7]. In this task participants perform 5 trials in each condition, which each included 4 chests (i.e. 20 memory targets per condition). Each chest has an item whose identity and location must be encoded, and then after a distractor phase recalled. Our version of the task is performed via head mounted display rather than on a desktop computer, with one condition being an augmented reality task in the real world, and the second condition a virtual reality task in a 3d recreation of the real environment which was performed in the same physical space (Fig 1). Our task was designed for the Meta Quest Pro using the Unity game engine. Condition order varied across participants to avoid order effects. Total experiment duration was ~1 hour. The experiment took place in a safe and quiet room. Our subjective measure was questionnaire answers on a Likert scale (1-7; 1 lowest, 7 highest). We asked participants about their feelings of subjective immersion,

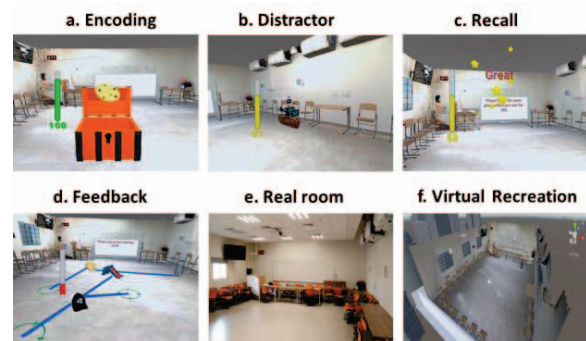


Figure 1: Experimental paradigm and environment. The task includes 4 stages - An (a) encoding stage, in which participants must locate chests and navigate to them, and then associate their contents with a specific location. (b) A distractor stage in which they needed to catch a ship. (c) A recall stage where participants were cued on each object and needed to select its location. (d) A feedback stage on their choice. (e) Image of the real room. (f) Image of the matched virtual recreation

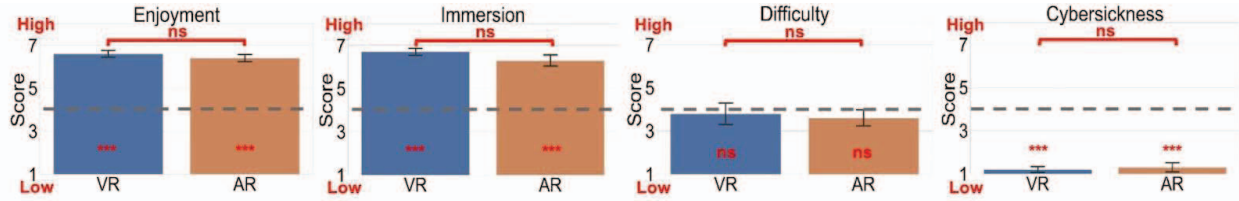


Figure 2: Subjective experience. Participants found both conditions to be significantly high in terms of enjoyment and immersion, of medium difficulty and reported low levels of cybersickness. There were no significant differences between the conditions in terms of enjoyment, Immersion, difficulty or cybersickness.

Enjoyment, difficulty and cybersickness. Our objective measure is the error distance - the distance between where participants thought the target was during recall and the object's actual location during encoding.

3 RESULTS

3.1.1 Subjective experience (Fig 2)

Immersion: Participants reported an immersion level of 6.7 ± 0.15 for VR and 6.3 ± 0.26 for AR (, “ $\pm$ ” denotes “Value $\pm$ SEM” throughout). Both of these scores indicate that participants found the experience immersive ($p < 2.7 \times 10^{-8}$, $p < 9.9 \times 10^{-6}$ respectively). These scores were not significantly different from each other ($p = 0.17$, paired t-test).

Difficulty: Participants reported a difficulty level of 3.8 ± 0.49 for VR and 3.6 ± 0.37 for AR. Both these scores indicate that participants found the experience not too difficult or too easy ($p = 0.69$, $p = 0.31$ respectively). These scores were not significantly different from each other ($p = 0.66$, paired t-test).

Enjoyment: Participants reported an enjoyment level of 6.6 ± 0.16 for VR and 6.4 ± 0.16 for AR. Both these scores indicate that participants found the experience enjoyable ($p < 6.7 \times 10^{-8}$, $p < 1.3 \times 10^{-7}$ respectively). These scores were not significantly different from each other ($p = 0.34$, paired t-test).

Cybersickness: Participants reported a cybersickness level of 1.2 ± 0.13 for VR and 1.3 ± 0.21 for AR. Both these scores indicate that participants did not find the experience to be nauseating ($p < 5.9 \times 10^{-9}$, $p < 4.9 \times 10^{-7}$ respectively). These scores were not significantly different from each other ($p = 0.68$, paired t-test).

3.1.2 Objective performance

Participants' average distance error was $1.03 \text{m} \pm 0.10$ in the VR condition and $1.00 \text{m} \pm 0.09$ in the AR condition (Fig 3). Both conditions demonstrated successful performance, significantly above chance ($p < 4.3 \times 10^{-9}$, $p < 5.6 \times 10^{-9}$, respectively). These scores were not significantly different from each other ($p = 0.71$).

4 DISCUSSION

We did not find any significant effects for reality modality in our spatial memory task. This finding is intriguing as reality modality and the challenges of disentangling it from other parameters are often mentioned as challenges for comparing performance across different types of extended reality tasks [3,6]. A question commonly posed to XR researchers is whether reality and the virtual experience are inherently different. Do we behave differently just by being inside virtual reality even if we are performing the exact same task? Our answer for the classic spatial memory task tested here is that apparently no, we treat them quite similarly.

Once our results are validated on a larger group of participants, for which data collection is currently ongoing, the next clear step will be to verify that this result is stable across subgroups of our participants by parameters such as gender, age, and amount of

experience with virtual reality. An additional important next step is to test direct transfer across these modalities - i.e. mix and match such that in some trials encoding is performed in one reality modality and recall in the other, a step for which data collection is now undergoing as well. If such transfer is indeed possible, it would be a beneficial basic science result that would support many applications for training virtually for real world tasks.

5 CONCLUSION

In conclusion, we found no significant differences between participant's performance in the VR and AR conditions for both subjective and objective measures. This suggests that when fully matched for movement, environment and interface type the effect of reality modality is not significant, which supports significant potential transfer across these modalities and potential uses of both AR and VR in general.

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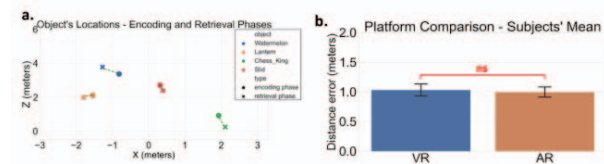


Figure 3: Objective performance. a. Our metric – the error distance between the target's location and the recalled location. b. No significant difference was found between spatial memory accuracy in the two conditions.